

Bayesian Estimation: Priors, Posteriors, MAP

Part 2: How much can one sample tell you about the world?

Probabilistic Machine Learning

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Outline

- 1 From prior to posterior
- 2 Point estimates
- 3 Predicting the next observation

Has the sun gone supernova?



Hypothetical scenario. AI-generated illustration, not to scale.

It is night. We cannot see the sun.

A sensor measuring radiation reports readings consistent with a supernova, but the sensor is noisy and can have elevated readings even in the absence of a supernova.

What should we conclude?

Last time: the frequentist view

Has the sun gone supernova?

To a frequentist, the sun has either gone supernova or it has not. That is a fact about the universe; no probability attaches to it. What is random is the measurement. Our trust in the measurement comes from how reliable they are in general, not from the particular reading we have.

Chapter 2 in these terms

θ^* is fixed and unknown. An estimator $\hat{\theta}(\mathbf{X})$ is a rule from data to a value, and its bias, variance, and mean squared error are averages over the data it could have seen.

The setup is xkcd 1132.

Today: the Bayesian view

Has the sun gone supernova?

A Bayesian does not know which universe she lives in: one in which the sun has gone supernova, or one in which it has not. She assigns each a probability before looking (supernovae are rare, so the first gets very little), and the sensor readings update those probabilities. Her conclusion is a probability for each universe, given the readings she actually has.

This chapter in these terms

Θ is a random variable. A prior $p(\theta)$ says what we believe before the data. Bayes' rule turns the prior and the data into a posterior $p(\theta|\mathbf{x})$. Every question about θ becomes a computation with the posterior.

The same readings, two questions: how reliable are my measurements (estimates), and how probable is each universe given what I saw.

The ingredients

Components

Data $\mathbf{x}$: a realization of $\mathbf{X}$, whose distribution depends on the parameter Θ . Θ is unknown, and therefore a random variable. Together they have a joint distribution $p_{\mathbf{X},\Theta}$, with marginals $p_{\mathbf{X}}$ and p_{Θ} .

Three steps

- 1 The prior $p_{\Theta}(\theta)$.
- 2 The data $\mathbf{x}$ and the likelihood $p_{\mathbf{X}|\Theta}(\mathbf{x}|\theta)$.
- 3 The posterior

$$p_{\Theta|\mathbf{X}}(\theta|\mathbf{x}) = \frac{p_{\Theta}(\theta) p_{\mathbf{X}|\Theta}(\mathbf{x}|\theta)}{p_{\mathbf{X}}(\mathbf{x})}, \quad \text{written } p(\theta|\mathbf{x}) = \frac{p(\theta)p(\mathbf{x}|\theta)}{p(\mathbf{x})}$$

once the argument makes the distribution clear.

Working up to a constant

Drop constants, then normalize

Drop every factor that is constant in θ first:

$$p(\theta|\mathbf{x}) \propto p(\theta)p(\mathbf{x}|\theta),$$

dropping constants from both $p(\theta)$ and $p(\mathbf{x}|\theta)$ too. Then normalize the right side.

How to normalize?

- 1 Recognize the distribution (today's route).
- 2 Integrate numerically.
- 3 Approximate: Markov chain Monte Carlo and variational inference, later in the course.

When only $\arg \max_{\theta} p(\theta|\mathbf{x})$ is wanted, no normalization at all.

$\propto$ is the working tool

Everything constant in θ goes, and the constant comes back at the end, if it is needed at all.

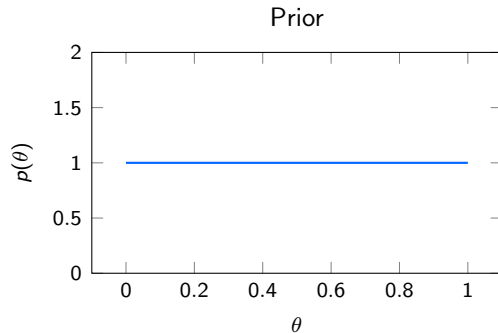
The traffic road: the prior

A road has light traffic on a day with probability θ . Over 10 days, 6 had light traffic. Three steps lead to the posterior.

Step 1: the prior

Before the data, nothing is known about Θ , so every value in $[0, 1]$ is equally plausible:

$$\Theta \sim \text{Uni}(0, 1), \quad p(\theta) = 1 \text{ on } [0, 1].$$



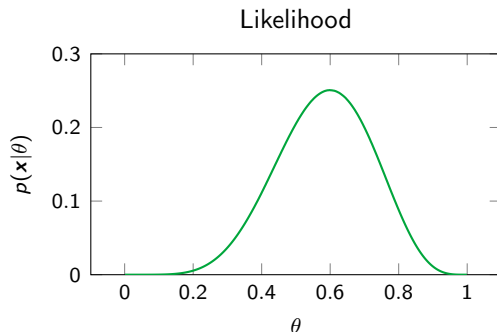
The traffic road: the likelihood

Step 2: the data and the likelihood

The data is $\mathbf{x} = (6, 4)$: 6 light and 4 heavy days. Given θ , the count of light days is $\text{Bin}(10, \theta)$, so

$$p(\mathbf{x}|\theta) = \binom{10}{6} \theta^6 (1 - \theta)^4.$$

Read as a function of θ , this is the likelihood. It peaks at $\theta = 6/10 = 0.6$, the MLE (last week's method); the binomial coefficient does not depend on θ .



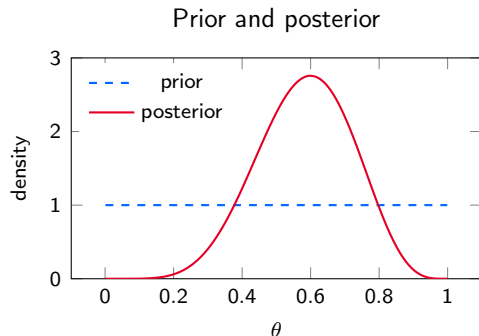
The traffic road: the posterior

Step 3: the posterior

$$p(\theta|\mathbf{x}) = \frac{p(\theta) p(\mathbf{x}|\theta)}{p(\mathbf{x})} = \frac{1 \cdot \binom{10}{6} \theta^6 (1-\theta)^4}{p(\mathbf{x})} \\ \propto \theta^6 (1-\theta)^4.$$

The grayed factors do not depend on θ and are dropped. As a function of θ , what is left is a Beta density up to its constant: $\Theta|\mathbf{x} \sim \text{Beta}(7, 5)$.

The mode is 0.6, the MLE; the mean is $7/12 = 0.583$; the standard deviation is 0.137.

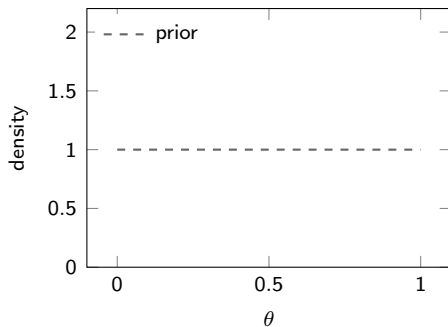


Your turn: one toss, then three

Let $X_i = 1$ denote a head and $X_i = 0$ a tail, with $X_i \mid \Theta = \theta \sim \text{Ber}(\theta)$. The prior on Θ is uniform.

Predict: sketch the posterior after a single head, before computing it.

- (a) Observe $X_1 = 1$. Find $p(\theta \mid X_1 = 1)$.
- (b) Observe $\mathbf{x} = (1, 1, 0)$. Find the posterior and $\Pr(\Theta > 1/2 \mid \mathbf{X} = \mathbf{x})$.



Your turn: one toss, then three (a)

(a) Observe $X_1 = 1$. Find $p(\theta \mid X_1 = 1)$.

Your turn: one toss, then three (a)

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Solution

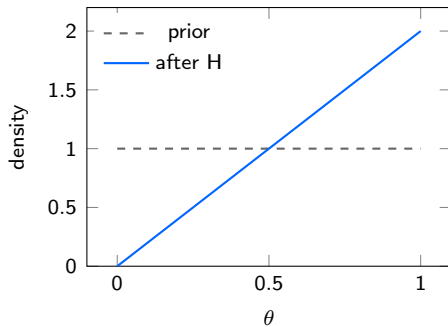
Prior: $p(\theta) = 1$, $\Theta \sim \text{Beta}(1, 1)$,

Likelihood: $p(X_1 = 1 \mid \theta) = \theta$,

Posterior: $p(\theta \mid X_1 = 1) \propto \theta = \theta^1(1 - \theta)^0$,

$\Theta \mid X_1 = 1 \sim \text{Beta}(2, 1)$,

$p(\theta \mid X_1 = 1) = 2\theta$.



Your turn: one toss, then three (b)

- (b) Observe $\mathbf{x} = (1, 1, 0)$. Find the posterior and $\Pr(\Theta > 1/2 \mid \mathbf{X} = \mathbf{x})$.

Your turn: one toss, then three (b)

- (b) Observe $\mathbf{x} = (1, 1, 0)$. Find the posterior and $\Pr(\Theta > 1/2 \mid \mathbf{X} = \mathbf{x})$.

Solution

Prior: $p(\theta) = 1, \quad 0 \leq \theta \leq 1,$

Likelihood: $p(\mathbf{x} \mid \theta) \propto \theta^2(1 - \theta),$

Posterior: $p(\theta \mid \mathbf{x}) \propto \theta^2(1 - \theta)$

$$= \theta^{3-1}(1 - \theta)^{2-1},$$

$$\Theta \mid \mathbf{x} \sim \text{Beta}(3, 2),$$

Your turn: one toss, then three (b)

(b) Observe $\mathbf{x} = (1, 1, 0)$. Find the posterior and $\Pr(\Theta > 1/2 \mid \mathbf{X} = \mathbf{x})$.

Solution

Prior: $p(\theta) = 1, \quad 0 \leq \theta \leq 1,$

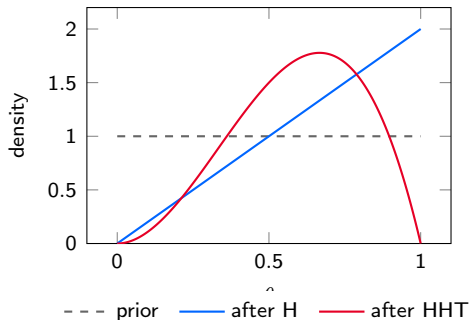
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 $= \theta^{3-1}(1 - \theta)^{2-1},$

$\Theta \mid \mathbf{x} \sim \text{Beta}(3, 2),$

Density: $p(\theta \mid \mathbf{x}) = 12\theta^2(1 - \theta),$

Tail: $\Pr(\Theta > \frac{1}{2} \mid \mathbf{x}) = 1 - F_{\text{Beta}(3,2)}(\frac{1}{2}),$
 $\Pr(\Theta > \frac{1}{2} \mid \mathbf{x}) = \frac{11}{16} = 0.6875.$



A head multiplies the curve by θ and tilts it right; a tail by $1 - \theta$ and tilts it left.

A Beta prior with Bernoulli data

Let $n = h + t$, where h is the number of heads and t the number of tails.

	Proportional form	Named distribution
Prior	$p(\theta) \propto \theta^{\alpha-1}(1-\theta)^{\beta-1}$	$\Theta \sim \text{Beta}(\alpha, \beta)$
Likelihood for data (h, t)	$p(\mathbf{x} \mid \theta) \propto \theta^h(1-\theta)^t$	$X \mid \Theta \sim \text{Bin}(n, \theta)$
Posterior	$p(\theta \mid \mathbf{x}) \propto \theta^{\alpha+h-1}(1-\theta)^{\beta+t-1}$	$\Theta \mid \mathbf{x} \sim \text{Beta}(\alpha + h, \beta + t)$

The observed counts add to the two Beta parameters.

Observed percentages can be misleading

A fraction θ of viewers like a video; the prior is uniform. Call the video high quality if $\theta > 0.8$.

Before the data: what is the prior probability that the video is high quality?

Observed percentages can be misleading

A fraction θ of viewers like a video; the prior is uniform. Call the video high quality if $\theta > 0.8$.

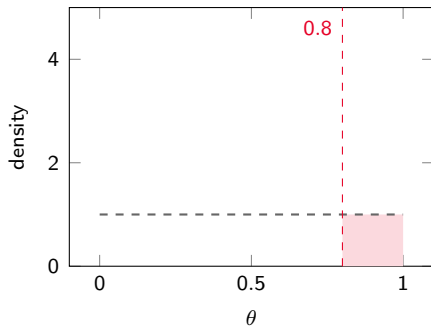
Before the data: what is the prior probability that the video is high quality?

After observing $\mathbf{x}$, use

$$\Pr(\Theta > 0.8 \mid \mathbf{x}) = \int_{0.8}^1 p(\theta \mid \mathbf{x}) d\theta.$$

Predict: which result makes you most confident that the video is high quality?

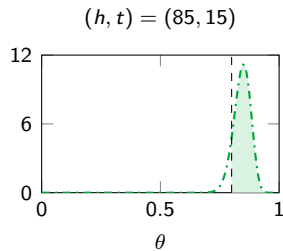
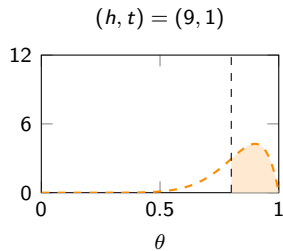
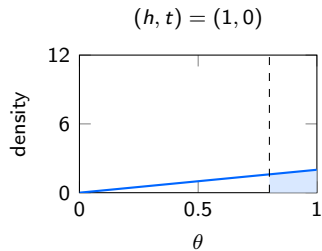
- (a) 1 like, 0 dislikes: 100% liked it.
- (b) 9 likes, 1 dislike: 90% liked it.
- (c) 85 likes, 15 dislikes: 85% liked it.



The posterior ranking reverses the observed percentages

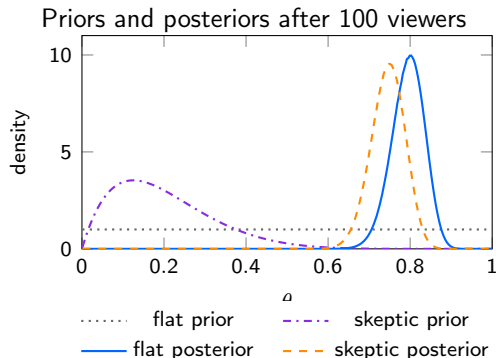
Under the uniform prior, $\Pr(\Theta > 0.8) = 1 - 0.8 = 0.20$.

Data (h, t)	Observed rate h/n	Posterior	$\Pr(\Theta > 0.8 \mid \mathbf{x})$
(1, 0)	1.00	Beta(2, 1)	0.36
(9, 1)	0.90	Beta(10, 2)	0.68
(85, 15)	0.85	Beta(86, 16)	0.88



Data reduces sensitivity to fixed Beta priors

Predict: after 100 viewers (80 likes, 20 dislikes), how far apart are the two means? After 1000?



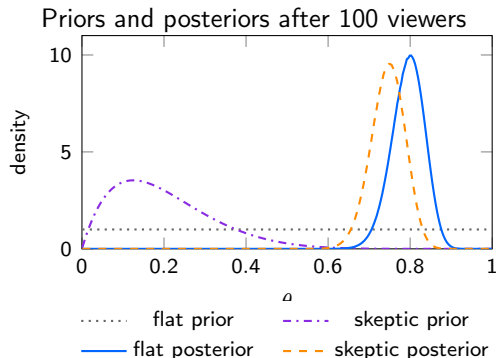
The two posterior means converge in this model

For fixed Beta priors with positive density near the data-generating value, the likelihood increasingly dominates as n grows.

Data reduces sensitivity to fixed Beta priors

Predict: after 100 viewers (80 likes, 20 dislikes), how far apart are the two means? After 1000?

viewers	flat prior	skeptic Beta(2, 8)	gap
5	0.71	0.40	0.31
20 (16 likes)	0.77	0.60	0.17
100 (80 likes)	0.79	0.75	0.05
1000 (800 likes)	0.80	0.79	0.01



The two posterior means converge in this model

For fixed Beta priors with positive density near the data-generating value, the likelihood increasingly dominates as n grows.

Two equivalent ways to update

Let $\mathbf{x}$ be the data already observed and $\mathbf{x}'$ the new data. Assume $\mathbf{X}' \perp\!\!\!\perp \mathbf{X} \mid \Theta$, so $p(\mathbf{x}' \mid \theta, \mathbf{x}) = p(\mathbf{x}' \mid \theta)$.

Use posterior as new prior

Update with the first batch:

$$p(\theta \mid \mathbf{x}) \propto p(\theta)p(\mathbf{x} \mid \theta).$$

Use the posterior as the prior for the second batch:

$$p(\theta \mid \mathbf{x}, \mathbf{x}') \propto p(\theta \mid \mathbf{x})p(\mathbf{x}' \mid \theta).$$

Pool the data with the original prior

Combine the two batches, then update once:

$$p(\theta \mid \mathbf{x}, \mathbf{x}') \propto p(\theta)p(\mathbf{x}, \mathbf{x}' \mid \theta).$$

Use conditional independence to factor the likelihood:

$$p(\mathbf{x}, \mathbf{x}' \mid \theta) = p(\mathbf{x} \mid \theta)p(\mathbf{x}' \mid \theta).$$

Both routes give the same posterior

$$p(\theta \mid \mathbf{x}, \mathbf{x}') \propto p(\theta)p(\mathbf{x} \mid \theta)p(\mathbf{x}' \mid \theta).$$

Shapes in the Beta family

$$p(\theta) \propto \theta^{\alpha-1}(1-\theta)^{\beta-1}.$$

For $\alpha, \beta \geq 1$, interpret the prior relative to the uniform Beta(1, 1) prior:

$$\underbrace{\alpha - 1}_{\text{prior successes}}, \quad \underbrace{\beta - 1}_{\text{prior failures}}.$$

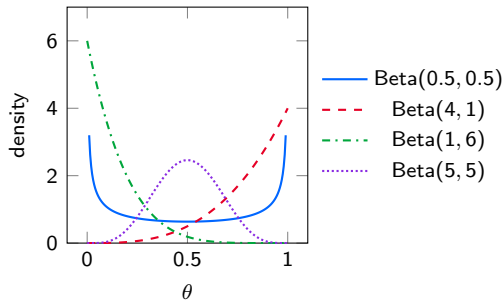
Total prior pseudocount: $\alpha + \beta - 2$.

A parameter below 1 has no pseudocount interpretation and can place substantial mass near a boundary.

Question: which curve fits each setting?

- (a) Little is known about traffic.
- (b) An allele causes a disease.

For more on choosing priors, read about *Jeffreys priors*.



A Poisson model for event counts

Let Y_i be the number of events in observation window i .

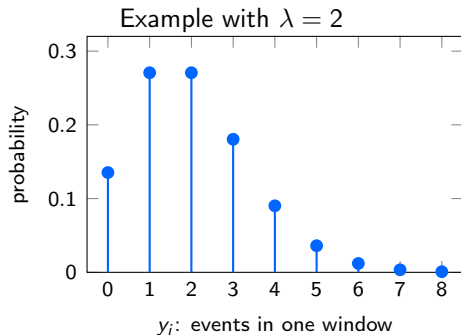
$$Y_i \mid \Lambda = \lambda \sim \text{Poi}(\lambda), \quad p(y_i \mid \lambda) = e^{-\lambda} \frac{\lambda^{y_i}}{y_i!}.$$

Here, λ is the expected count in one window:

$$\mathbb{E}[Y_i \mid \lambda] = \lambda, \quad \text{Var}(Y_i \mid \lambda) = \lambda.$$

Why this model can make sense

The observations are counts in comparable windows. Events occur independently at a roughly stable rate within each window.



The same recipe for counts: Poisson data, Gamma prior

For n independent windows, let $S = \sum_{i=1}^n y_i$. The Gamma distribution uses the shape–rate parameterization.

	Proportional form	Named distribution
Prior	$p(\lambda) \propto \lambda^{\alpha-1} e^{-\beta\lambda}$	$\Lambda \sim \text{Gamma}(\alpha, \beta)$
Likelihood for data y_1^n	$p(y_1^n \lambda) \propto \lambda^S e^{-n\lambda}$	$Y_i \Lambda \sim \text{Poi}(\lambda)$
Posterior	$p(\lambda y_1^n) \propto \lambda^{\alpha+S-1} e^{-(\beta+n)\lambda}$	$\Lambda y_1^n \sim \text{Gamma}(\alpha + S, \beta + n)$

S adds to the Gamma shape, and n adds to the Gamma rate.

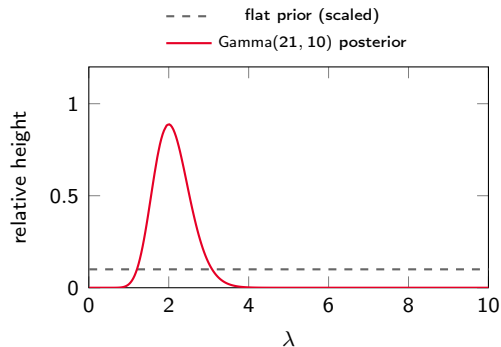
A convenient flat prior for a rate

Data	$n = 10, S = 20$
Reference prior	$p(\lambda) \propto 1, \lambda > 0$
Posterior	$\Lambda \mid y_1^{10} \sim \text{Gamma}(21, 10)$
Posterior mean	$\mathbb{E}[\Lambda \mid y_1^{10}] = 2.1$

Improper prior, proper posterior

$\int_0^\infty 1 d\lambda = \infty$, but the posterior density integrates to one.

“Flat” depends on the parameterization.



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Point estimates depend on the loss

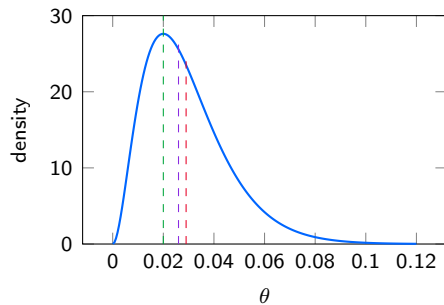
Sometimes we need a single number to summarize the posterior distribution.

The choice depends on the loss function.

Posterior loss	Best estimate
$\mathbb{E}[(\Theta - \hat{\theta})^2 \mathbf{x}]$	posterior mean $\mathbb{E}[\Theta \mathbf{x}]$
$\mathbb{E}[\Theta - \hat{\theta} \mathbf{x}]$	posterior median
discrete 0–1 loss	posterior mode

For continuous Θ , MAP maximizes the posterior density:

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} p(\theta | \mathbf{x}), \quad \hat{\theta}_{\text{mean}} = \mathbb{E}[\Theta | \mathbf{x}].$$



mode 0.020 < median 0.026 < mean 0.029

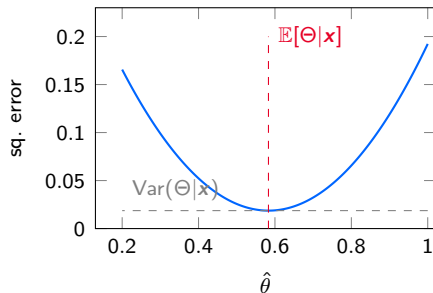
Why the posterior mean minimizes the squared error

With $\bar{\theta} = \mathbb{E}[\Theta | \mathbf{x}]$,

$$\mathbb{E}[(\hat{\theta} - \Theta)^2 | \mathbf{x}] = \text{Var}(\Theta | \mathbf{x}) + (\hat{\theta} - \bar{\theta})^2.$$

The cross term is zero because $\mathbb{E}[\Theta - \bar{\theta} | \mathbf{x}] = 0$.

The loss is minimized at $\hat{\theta} = \bar{\theta}$.



The floor is the posterior variance

Minimum posterior MSE = $\text{Var}(\Theta | \mathbf{x})$.

Mean, median, and mode for a Beta posterior

Let $\Theta \mid \mathbf{x} \sim \text{Beta}(a, b)$, where $a = \alpha + h$ and $b = \beta + t$.

Summary	Formula
Mean	$\frac{a}{a+b}$
Median	$\frac{a - 1/3}{a + b - 2/3}$ (approx.)
Mode	$\frac{a - 1}{a + b - 2}$

The median approximation and the mode formula assume $a, b > 1$.

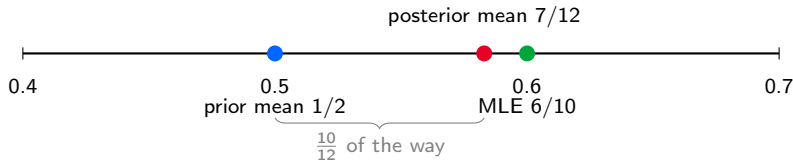
Uniform prior

$$\hat{\theta}_{\text{MAP}} = \hat{\theta}_{\text{MLE}} = h/n.$$

Between the prior and the data

The posterior mean is a weighted average of the MLE and the prior mean:

$$\frac{h + \alpha}{n + \alpha + \beta} = \frac{n}{n + \alpha + \beta} \cdot \frac{h}{n} + \frac{\alpha + \beta}{n + \alpha + \beta} \cdot \frac{\alpha}{\alpha + \beta}.$$



Data weight n , prior weight $\alpha + \beta$

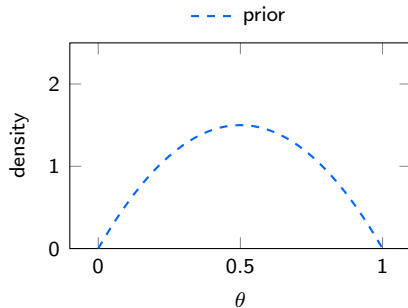
As n grows, the posterior mean moves toward the MLE.

Your turn

Given: $n = 3$, $h = 2$, $t = 1$, and $\Theta \sim \text{Beta}(2, 2)$.

- (a) The posterior?
- (b) Posterior mean, posterior mode, the MLE, the prior mean?
- (c) Check the weighted average.

Solution



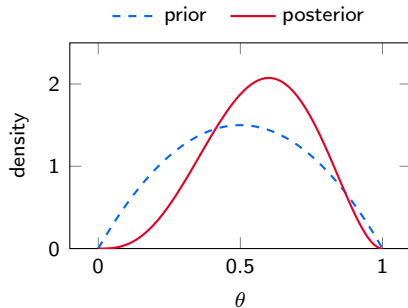
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Solution

(a) $p(\theta | \mathbf{x}) \propto \theta^3(1 - \theta)^2$, $\Theta | \mathbf{x} \sim \text{Beta}(4, 3)$.



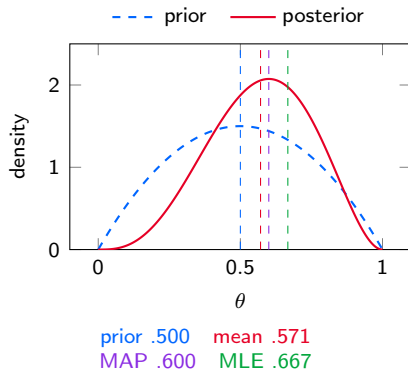
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Solution

- (a) $p(\theta | \mathbf{x}) \propto \theta^3(1 - \theta)^2$, $\Theta | \mathbf{x} \sim \text{Beta}(4, 3)$.
- (b) mean = $4/7$, MAP = $3/5$, MLE = $2/3$, prior mean = $1/2$.



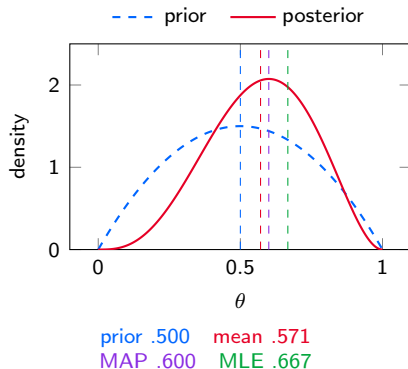
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- (c) Check the weighted average.

Solution

- (a) $p(\theta | \mathbf{x}) \propto \theta^3(1 - \theta)^2$, $\Theta | \mathbf{x} \sim \text{Beta}(4, 3)$.
- (b) mean = $4/7$, MAP = $3/5$, MLE = $2/3$, prior mean = $1/2$.
- (c) $\frac{4}{7} = \left(\frac{3}{7}\right) \left(\frac{2}{3}\right) + \left(\frac{4}{7}\right) \left(\frac{1}{2}\right)$.



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The posterior predictive distribution

Two distributions are already available:

$$\underbrace{p(\theta \mid y_1^n)}_{\text{posterior}} \quad \underbrace{p(y_{n+1} \mid \theta)}_{\text{sampling model}} .$$

Average the sampling model over the posterior:

$$p(y_{n+1} \mid y_1^n) = \int p(y_{n+1}, \theta \mid y_1^n) d\theta = \int p(y_{n+1} \mid \theta) p(\theta \mid y_1^n) d\theta.$$

For Bernoulli data,

$$\Pr(Y_{n+1} = 1 \mid y_1^n) = \mathbb{E}[\Theta \mid y_1^n].$$

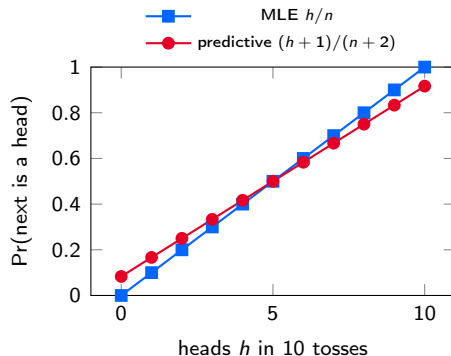
Beta–Bernoulli predictive probability

$$\Pr(Y_{n+1} = 1 \mid y_1^n) = \mathbb{E}[\Theta \mid y_1^n] = \frac{h + \alpha}{n + \alpha + \beta}.$$

General Beta prior $\frac{h + \alpha}{n + \alpha + \beta}$

Uniform prior $\frac{h + 1}{n + 2}$

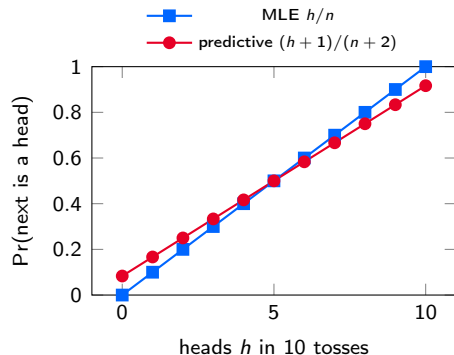
For $\alpha, \beta > 0$, the predictive probability lies strictly between 0 and 1 for every finite sample.



Extreme counts and prior smoothing

Ten tosses with a uniform Beta(1, 1) prior:

Heads	MLE h/n	Predictive $(h+1)/(n+2)$
$h = 0$	0	1/12
$h = 10$	1	11/12



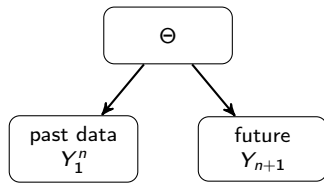
The uniform prior smooths the observed proportion

Finite-sample predictions stay away from the endpoints 0 and 1.

The expected value of the next observation

$$\mathbb{E}[Y_{n+1} \mid y_1^n] = \mathbb{E}_{\Theta \mid y_1^n}[\mathbb{E}(Y_{n+1} \mid \Theta)].$$

First compute the conditional mean at each θ . Then average those means using the posterior distribution of Θ .



$Y_{n+1} \perp\!\!\!\perp Y_1^n \mid \Theta$: once Θ is known, the past does not change the distribution of the next observation.

Poisson: the expected count tomorrow

For counts $Y_i \mid \Lambda \sim \text{Poi}(\Lambda)$ and posterior $\Lambda \mid y_1^n \sim \text{Gamma}(\alpha + n\bar{y}, \beta + n)$, the expected count tomorrow follows from iterated expectations:

$$\begin{aligned}\mathbb{E}[Y_{n+1} \mid y_1^n] &= \mathbb{E}_{\Lambda \mid y_1^n}[\mathbb{E}(Y_{n+1} \mid \Lambda, y_1^n)] && \text{(law of iterated expectations)} \\ &= \mathbb{E}_{\Lambda \mid y_1^n}[\mathbb{E}(Y_{n+1} \mid \Lambda)] && (Y_{n+1} \perp\!\!\!\perp Y_1^n \mid \Lambda) \\ &= \mathbb{E}_{\Lambda \mid y_1^n}[\Lambda] && (Y_{n+1} \mid \Lambda \sim \text{Poi}(\Lambda) \implies \mathbb{E}[Y_{n+1} \mid \Lambda] = \Lambda) \\ &= \frac{\alpha + n\bar{y}}{\beta + n} && \text{(mean of Gamma}(\alpha + n\bar{y}, \beta + n)\text{)}.\end{aligned}$$

Weighted average of prior mean and sample mean

$$\mathbb{E}[Y_{n+1} \mid y_1^n] = \frac{\beta}{\beta + n} \left(\frac{\alpha}{\beta} \right) + \frac{n}{\beta + n} \bar{y}$$

As $n \rightarrow \infty$, the weight on the prior fades and the prediction approaches $\bar{y}$.

- Posterior $\propto$ prior $\times$ likelihood.
- Bernoulli counts update a Beta distribution. Poisson counts update a Gamma distribution.
- Sufficient statistics add to the prior parameters.
- The loss function determines the appropriate point estimate.
- Prediction averages the sampling model over the posterior.

Next: Gaussian prior and likelihood, with frequentist and Bayesian estimators side by side.